O-B-491d
May 31, 1968
SUPERSEDING
Fed. Spec. O-B-491c
September 10, 1964

FEDERAL SPECIFICATION

BLUING, LAUNDRY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type of liquid laundry bluing (nonsour, sour) for general laundry use (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

Federal Specifications:

CCC-C-430 - Cloth, Cotton, Sheeting.
CCC-T-191 - Textile Test Methods.
PPP-B-636 - Box, Fiberboard.
PPP-C-186 - Container, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The bluing shall consist of a solution of a water-soluble, aniline blue dye. It shall be free from adulterants and shall not be fast to laundering. When diluted with water in the proportions recommended by the supplier, the bluing shall yield a stock solution which, when applied under the usual conditions of laundry operation, shall tint white goods to a bright white.

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3.2 Acidity (pH). The pH value of the bluing shall be not less than 5.5 and not greater than 8.5 when tested as specified in 4.3.2.

3.3 Insoluble dye. There shall be no evidence of insoluble dye when tested as specified in 4.3.2.

3.4 Presence of aniline blue dye and absence of ultramarine blue, prussian blue and methylene blue. The bluing, when tested as specified in 4.3.2, shall show the presence of an aniline dye and absence of ultramarine blue, prussian blue or methylene blue.

3.5 Resistance to decomposition by acid and alkali. No precipitate or sediment shall be formed in acid, neutral and alkaline solutions of the bluing when tested as specified in 4.3.2.

3.6 Fastness to ironing. There shall be no appreciable fading nor yellowing on ironing of blued fabric, when tested as specified in 4.3.2.

3.7 Ease of removal in subsequent washing. The bluing shall be removed from cotton fabric by laundering when tested as specified in 4.3.2. The blued, washed fabric shall be equal in whiteness to the untreated control fabric.

3.8 Color intensity of acid, neutral, and alkaline bluing solutions. The bluing, when tested as specified in 4.3.2, shall give transmission values not greater than 30 percent in acid solution, not greater than 45 percent in neutral solution, and not greater than 55 percent in alkaline solution. The percent transmission of the acid solution shall be no greater than that of the neutral solution. Based on the percent transmission readings obtained in the color intensity test (see 4.4.8), the alkaline solution shall show a percent decrease in color intensity not greater than 22 percent in comparision to the color intensity of the neutral solution.

3.9 Directions for use. Each glass or plastic bottle of laundry bluing shall be durable and legibly marked with the following directions for use:

LIQUID LAUNDRY BLUING

(NON-SOUR, SOUR)

"WARNING: NOT TO BE USED ON BABIES DIAPERS"

Directions. Prepare a stock solution of the bluing by diluting 1/ fluid ounces of the concentrated blue with 5 gallons of fresh water. Rinse the container used to measure the blue with some of the water and add the rinsings to the stock solution. Store the stock solution in glass, enameled, stainless steel, or glazed earthenware containers; do not store in iron or galvanized metal containers.

Blue is added to the wash-wheel as follows: (1) Use 2 ounces of blue stock solution per 50-pound load; (2) put the proper amount of blue stock solution in a pail and add water to dilute it further; (3) add this pail of blue to the load along the length of the washer while the cylinder, as seen through the supply door, is going down. If the bluing results are too light or too dark, the operator may vary the amount of stock solution used to obtain the desired tint.

3.10 Workmanship. The finished product shall be clean and uniform.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

1/Manufacturer shall insert here the number of fluid ounces (to the nearest 1/2 ounce) of his product which is to be used in preparing 5 gallons of stock solution.

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4.2.1 Inspection of materials and components. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for the defects in the applicable subparagraphs at inspection levels and acceptable quality levels (AQLs) set forth in 4.3.1.4. Random samples shall be drawn from each lot for examination of visual, net contents, workmanship and preparation for delivery defects. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled bottles of the specified capacity for examination in 4.3.1.1 and 4.3.1.2 and in units of shipping containers for examination under 4.3.1.3.

4.3.1.1 Examination for visual defects. The sample unit for this examination shall be the contents of one filled bottle.

Examine	Defect
Container (for construction)	Not in container specified. Any leakage of contents. Any hole, tear, puncture, scuff, dent. Seals or closures not as specified.
Markings (label or printing)	Omitted, illegible, incorrect, incomplete, not in accordance with requirements. Label or printing not clear, smeared, ink not as specified. Directions for use or precautionary markings not as specified (see 3.9).
Workmanship	Not in liquid form. Not clean; not uniform.
Color	Not blue.

4.3.1.2 Examination for net contents. The sample unit for this examination shall be one filled bottle. The lot shall be unacceptable if the average net content per bottle is less than the specified or indicated quantity.

4.3.1.3 Examination of preparation for delivery. An examination shall be made to determine that material, contents, and markings comply with the requirements of section 5. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Closed shipping containers shall be examine for closure defects. The lot size shall be the number of shipping containers in the end item inspection lot.

Examine	Defect
Markings (exterior and interior)	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence, or method of application (see 5.3).
Material	Any nonconforming component; component missing, damaged, or otherwise defect. Inadequate application of components; such as incomplete closure of container flaps, loose or inadequate sealing, strapping or stapling.
Contents	Number of interior packages per shipping container is more or less than specified. Bulged or distorted container. Number of primary containers per intermediate package is less than specified.

4.3.1.4 Inspection levels and acceptable quality levels (AQLs) for examination. The inspection levels, for determining the sample size, and the acceptable quality levels (AQLs), expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection levels	AQLs
4.3.1.1	S-3	2.5
4.3.1.2	S-2	N/A
4.3.1.3	S-2	4.0

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4.3.2 Testing of the end item. The end item shall be tested for the characteristics specified in table I. The lot size shall be expressed in quarts of liquid bluing. The sample unit for testing shall be a two-quart composite obtained by combining equal portions from samples selected at random through out the lot. The number of portions of products to make up the composite shall be as indicated below. There shall be no failure of any test to meet the specified requirements.

Lot size (quarts)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE I. Testing of the end item

Characteristic	Specification reference		Requirements applicable to		Number determinations per unit	Results reported as		Inspect level	AQL
	Requirement	Test method	Individual unit	Lot aver		Pass or fail	Numerically to nearest		
Material	3.1	1/							
Acidity (pH)	3.2	4.4.2			Average of 2		0.1 pH		
Insoluble dye	3.3	4.4.3			1	X			
Presence of aniline blue dye and absence of ultramarine, prussian and methylene blue	3.4	4.4.4			1	X			
Resistance to decomposition by acid and alkali	3.5	4.4.5			1	X			
Fastness to ironing	3.6	4.4.6			1	X			
Ease of removal in washing	3.7	4.4.7			1	X			
Color intensity of acid, neutral, and alkaline bluing solutions	3.8	4.4.8			1	X			

1/A certificate of compliance will be furnished the contracting officer for this characteristic.

4.4 Test methods.

4.4.1 Stock items.

4.4.1.1 Stock solution. One thousand ml. of stock solution shall be prepared by diluting the bluing with distilled water in the proportions recommended by the manufacturer (see 3.9).

4.4.1.2 Stock cloth. Sheeting conforming to CCC-C-430 shall be laundered and then ironed on a flat-bed press as specified in method 5550 of CCC-T-191.

4.4.2 Acidity (pH). The pH value of the sample, as received, shall be determined at 25° ± 2° C, using a pH meter having a sensitivity and readibility of 0.1 pH unit. A sealed type glass electrode shall be used in conjunction with a potassium chloride-calomel reference electrode. The pH meter shall be calibrated before use against a standard buffer solution of pH7.

4.4.3 Insoluble dye. The sample, as received, shall be thoroughly mixed by shaking and 10 ml. of the solution shall be added by means of a pipette, into a 1,500 ml. beaker, diluted with 1,000 ml. of distilled water and stirred. The solution shall be filtered through a Whatman No. 12 filter paper (32.0 cm.). The filter paper shall be visually observed for deposits of insoluble matter. (Readily and completely soluble dyes color the paper uniformly.)

4.4.4 Presence of aniline blue dye and absence of ultramarine, prussian and methylene blue. Thirty-five ml. of distilled water shall be added to each of four 75 ml. test tubes containing 5 ml. of stock solution. One of these test tubes shall be retained as a control. To the others shall be added 10 ml. of 1:1 hydrochloric acid, 10 ml. of 10 percent sodium hydroxide, and 10 ml. of concentrated ammonium hydroxide, respectively. The solution shall be mixed well and the color changes

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compared with table II.

TABLE II. Color comparison

Reagent	Aniline dye	Ultramarine blue	Prussian blue	Methylene blue
1:1 HCl	Turns slightly darker in color or turns emerald green.	Decolorized	No action	Turns amber
10 percent NaOH	Turns purple or no action.	No action	Decolorized	Turns reddish brown
NH_4OH (27 percent NH_3)	Turns lighter	No action	Partially decolorized	Turns reddish brown

4.4.5 Resistance to decomposition by acid and alkali.

4.4.5.1 Reagents.

Sodium hydroxide (NaOH) prepared carbonate free and standardized according to usual volumetric procedure.

Potassium acid phthalate ($KHC_8H_4O_4$) - molecular weight 204.2 - ACS reagent grade is of sufficient purity.

Monopotassium acid phosphate (KH_2PO_4) - molecular weight 136.2 - ACS reagent grade shall be recrystallized twice from water and dried in oven at 110° to 120° C.

Boric acid (H_3BO_3) - molecular weight 62.0 - ACS reagent grade shall be recrystallized and dried in air.

Buffer solutions shall be prepared as follows:

pH	Buffer solution composition
4.0	0.40 ml. 0.1N NaOH + 50 ml. 0.1M $KHC_8H_4O_4$ + distilled water to 100 ml.
7.0	29.63 ml. 0.1N NaOH + 50 ml. 0.1M KH_2PO_4 + distilled water to 100 ml.
8.0	3.97 ml. 0.1N NaOH + 50 ml. 0.1M H_3BO_3 + distilled water to 100 ml.

4.4.5.2 Procedure. Two ml. of stock solution shall be added, by means of a pipette, to each of three 250 ml. beakers containing 100 ml. of buffer solution having a pH of 4.0, 7.0, and 8.0, respectively. The buffer solution with bluing shall be stirred well and allowed to stand for 24 hours. The solution shall be observed for the presence of precipitation or sediments.

4.4.6 Fastness to ironing. A specimen of stock cloth approximately 8 by 8 inches in dimension shall be prepared and soaked for 5 minutes, at 90° ± 5° F., in 1,000 ml. of distilled water to which has been added 15 ml. of stock solution. The cloth shall be removed and passed once through the wringer of a washing machine (do not wring by hand), then dried in air. The specimen shall be cut in half and one half shall be ironed as specified in method 5550 of CCC-T-191. Using the other half as a control, color comparisons shall be made by visual observation to determine whether the sample has faded appreciably or turned yellow.

4.4.7 Ease of removal in subsequent washing. Two specimens of stock cloth about 2-1/2 by 4 inches in dimensions shall be cut. One specimen shall be soaked for 5 minutes, at 90° ± 5° F., in 1,000 ml. of distilled water to which has been added 15 ml. of stock solution. The cloth shall then be dried in air. The other specimen of cloth shall be used as a control. Both specimens shall be washed separately, as specified in method 5610 of CCC-T-191. The bluing shall be removed completely from the sample cloth as determined by comparison with the control specimen.

4.4.8 Color intensity of acid, neutral, and alkaline bluing solution.

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4.4.8.1 Apparatus.

Spectrophotometer. The spectrophotometer shall be used with a 550 millimicron wave length setting, a PC-4 glass filter, and 13 by 13 by 105 millimeter (mm.) square cuvette. It shall be adjusted to read 100 percent transmission on the galvanometer using distilled water.

4.4.8.2 Procedure. Three hundred ml. of a pH 4.0 buffer solution (see 4.4.5 for formula) shall be added to a 400 ml. beaker. By means of a pipette or burette, add exactly 5 ml. of stock solution to the beaker, stir well, and allow to stand for exactly 5 minutes. At the end of 5 minutes, the color intensity (percent transmission) of the solution shall be measured with the previously standardized instrument. The measurement shall be repeated using buffer solutions of pH 7.0 and 8.0. Percent decrease in color intensity shall be calculated as follows:

$$\text{Percent decrease in color intensity} = \frac{T_{alk} - T_n}{T_n} \times 100$$

Where:

T_{alk} is percent transmission of alkaline solution.
T_n is percent transmission of neutral solution.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. One quart or one gallon of bluing, as specified (see 6.2), shall be packaged in a plastic bottle conforming to group A, class 2, style 1, closure A, seal A of PPP-C-186.

5.1.2 Level C. Bluing shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Twenty-four 1-quart or six 1-gallon bottles of bluing, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be provided with interlocking fiberboard partitions providing an individual cell for each bottle, and top and bottom fiberboard pads conforming to grade V3c of PPP-F-320. Top and bottom pads shall be cut 1/4 inch less than the full inside length and width of the shipping container. The height of the interlocking partitions shall be equal to the inside depth of the container less the combined thickness of the top and bottom pads. Closure, waterproofing by means of tape, and reinforcing with strapping or tape banding shall be in accordance with the appendix of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Civil agencies. Twelve 1-quart or six 1-gallon bottles of bluing, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, class domestic, grade 200 of PPP-B-636. Each shipping container shall be provided with top and bottom fiberboard pads cut 1/4 inch less than the full inside length and width of the shipping container. The pads shall be fabricated from the same material as the shipping container. Closure shall be in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.2 Military requirements. Twenty-four 1-quart or six 1-gallon bottles of bluing, packaged as specified in 5.1, shall be packed as specified in 5.2.2.1, and in addition shall be provided with interlocking fiberboard partitions providing an individual cell for each bottle. The partitions shall be fabricated from the same material as the shipping container and their height shall be equal to the inside depth of the container less the combined thickness of the top and bottom pads.

5.2.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c or V3s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the container specification. The shipping container material may also be grade V4s of PPP-F-320.

5.2.3 Level C. Bluing, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

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5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.3 Labeling. Each bottle of liquid bluing shall be durably and legibly marked with the instructions for use as specified in 3.9.

6. NOTES

6.1 Intended use. The laundry bluing covered by this specification is intended for general laundry use to neutralize the natural slightly yellow shade of fabrics by coloring them with a faint blue tint. The bluing works satisfactorily under acid, alkaline, or neutral conditions and may therefore be used whether or not souring is done.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Whether bluing shall be packaged in 1-quart or 1-gallon bottles (see 5.1.1).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.2.1).

6.3 Basis for award. The basis for award should be determined by the following formula:

$$\text{(Delivery price per quart)} \times \frac{A}{5}$$

Where:

A = number of fluid ounces of bluing recommended to prepare 5 gallons of stock solution (see 3.9).

6.4 Standard Pack (Civil Agencies). The Standard Pack for this commodity shall be as specified in 5.1.1 and 5.2.2.1. The Standard Pack requirements are intended for use in procurement of stores stock replenishments. Procuring officers should use the Standard Pack requirements when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot, or immediate storage point for temporary storage, subsequent issue, and shipment to evantual user.

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Army - GL
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Review activities:

Army - GL, MD
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Bluing, Laundry - O-B-491d

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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